Carnegie Mellon University
Computer Science Academy

Description

Describe your program. What do you envision your app will do?

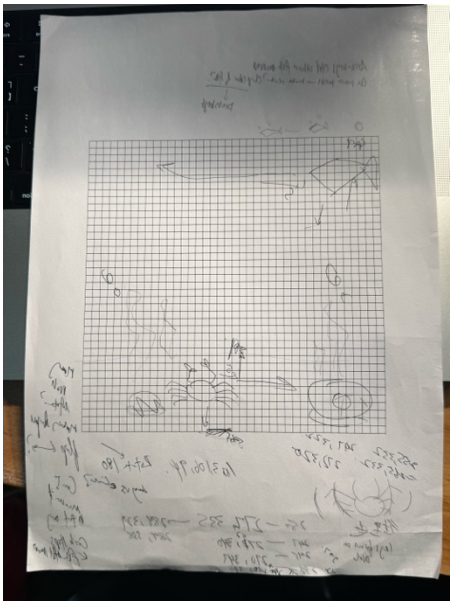
Note: At this step, do not be concerned about how to do it, just focus on what you envision.

I envision that it'll when the fish moves (left right), the crab will also move but opposite direction – some goldfish eat crabs. When it reaches the hiding areas ("hide" and "rock"), it'll disappear/get covered up. The fish and crab will stay on the screen (range of 0 to 400 for x).

Canvas Design

Sketch your canvas. Label any animation or user interaction.

- **Colors:** What colors will be dominant in your app? Are there specific shades that are important to the app's functionality or aesthetic?
- **Animations/Movements:** Will there be any animated elements? If yes, describe how you envision them. For example, will a shape move across the screen?



The crab and fish will move across screen in opposite directions. Fish will be main thing following arrow keys, crab will go opposite.

A somewhat faded backdrop with a vivid fish.

Code Design

Plan your code. List here the coding concepts and properties you plan to use to accomplish your goals, and why you plan to use them. These should incorporate at least two new concepts from this unit. Briefly describe what purpose each element will serve in your project. Aim for at least four to six properties.

Concept	Purpose

def onKeyPress	"Trigger" the movement of the fish and crab to do what I think it will do.
If and elif	Direct which direction crab and fish will go – only left and right. Elif so that nothing happens when other keys are pressed.
(the "ifs" in if and elif)	Has the distance the fish and crab will travel, as well as making sure that fish crab won't go past the x=0 and x=400 line. Also setup for the containsShape line so it'll work.
containsShape	Makes it so that crab will disappear when gotten to that shape.
Group	Make it neater and easier to move together in the moving parts.

Creative Task Reflection

After completing your creative task use this form to reflect on your work.

Difficulties and Opportunities

For this section, go in-depth into either a difficulty you faced or an opportunity you saw to improve or add to your code.

If you Faced a Difficulty:

- Describe the Problem: What specific issue did you encounter?
- Attempted Solutions: What did you try in order to solve the problem?
- Final Solution: What was the final solution that fixed the issue?

If you Found an Opportunity:

- Reason for the Addition/Change: What made you consider this opportunity? Was it a challenge or an aha-moment?
- Impact on Design: How did seizing this opportunity impact other aspects of your design?
- Design Trade-offs: What aspects of your original design did you have to adjust or remove to incorporate this new feature?

A difficulty would be making the crab stop moving when it got to the hide and rock. Originally, I was just going to have it reach the canvas edge and stop, but realized why have a hide and do nothing about it? I wanted to make it that when the crab reached it, It will disappear.
I first did it as "if (hide.containShape == True)" then it stops. I remembered there was a similar exercise from earlier, so I tried "coping" that with stopping the entire app. It stopped the entire thing. Tried

something else, and there would be an error or it'll go past that limit (there was also a time where I forgot what containShape was called). So then I asked for help and saw that it could be written the other way around – instead of making it containShape == True and it stops or disappears, make it containShape == False and move the X value. That worked, and I did it again for the other hide, which was going to be a hole. That was another issue.
The counterintuitive way worked, but for this the hole tunnel wasn't as big as the crab so it won't be completely covered (didn't help that my opacity was about 50). I then tried hitShape and made visible false, which worked, but when moving away from the shape crab didn't appear again until after a minute of spamming arrow keys. And it proceeded to disappear again. That was annoying. I ended up just making it a rock instead and did the containShape again. Not the most creative or happiest thing that day.

Finished Project Code Outline

List **only one instance** of each important coding concept used in your project, along with the corresponding line number, and describe how it contributes to the overall function of your program. **Highlight the aspect you are most proud of.** For each concept, also include a brief definition to demonstrate your understanding. By focusing on one instance of each concept, aim to provide more depth and clarity in your responses.

Concept	Line number(s)	Definition	Purpose
If (hide.containsShape(crab) == False):	64	This makes it so that as long as the hide isn't covering the crab the crab will keep moving.	This is for when the crab reaches the hiding place then it won't move after it (since it's reached its goal).
if (fish.rotateAngle == 180):	66	This is so that when the fish is 180 degrees rotated, then some "thing" else will happen.	So that the fish won't be swimming backwards when I press the left arrow key, and so when I press left arrow it'll spin around and face forward while moving.
elif (key == 'left'):	72	Only move that direction if only the left key is pressed.	For the fish, it's going to move left and up, so that it'll reach the starting place. While for the crab it's going to move right until it reaches the hiding place other there.
if (fish.centerX >= 70):	78	Limiting the domain of the fish, makes it so that its center doesn't pass the line/x value of 70.	By limiting how far the x value the fish travels, it's so it won't go off the screen and will stay inside the canvas.

Future goals

Reflect on your entire design and coding process, from conceptualization to implementation. In this section, consider not only what you would change in the final product but also how you would approach the entire project differently.

- **Code Changes:** What specific changes would you make to your code if you had more time or if you were to start from scratch? Would you add new features or refine existing ones?
- **Process Adjustments:** Did you stick to the original plan, or were there deviations? Were these changes beneficial or did they lead to complications? How could your planning or approach be improved?
- **Resource Utilization:** Did you make the best use of available resources, like class notes, tutorials, or peer help? What additional resources could have been beneficial?
- **Time Management:** Reflect on how effectively you managed your time. What would you do differently to ensure a smoother process?

By considering these aspects, you will not only think about 'what' you would change but also 'how' and 'why,' which can be incredibly enlightening for future projects.

I would add in other arrow keys/keys to made it more interesting, might be an up/down, press “b” and bubbles appear, when fish is on contact with crab the crab disappears. Or maybe as fish move bubbles trail after (above) it, move one arrow there’s a bubble, move again there’s another, with the maximum of bubbles co-existing being two. Something else might be when the crab moves, the legs also move, and when the fish moves the tail “swishes”. Maybe the fish’s eyes can follow the crab’s movement and when the crab’s hiding the eye snaps back to the normal place (can fish detect smell?). As well as make the tunnel hiding place actually work instead of being lazy and changing the idea.
As for the process, I mostly stuck to the original idea – the fish, crab, hide, seaweed, move left/right – but I also added in another hiding place, and that when the fish moves vertically, there’s also an action of moving down (going right) and up (left, to starting position). These changes made it so that something else happened other than running around.
For resource utilization, I used notes, past exercises such as one where it stopped after a certain point, and asked for help for the stop moving when it reached the hide.
I think I did better with time management with this one. But I should’ve spent less time looking at crabs.

Scroll down for submission checklist.

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Pre-Submission Checklist for Creative Tasks

Program Design

- [ ] Described what your program will do clearly.
- [ ] Incorporated at least two concepts from the current unit.
- [ ] Added annotations to explain colors, movements, and user interactions.

Canvas Design

- [ ] Sketched your canvas layout.
- [ ] Labeled any animation or user interaction on the canvas sketch.
- [ ] Detailed dominant colors and their importance.
- [ ] Described any animations or movements.

Code Design

- [ ] Listed at least four to six coding concepts or properties you plan to use.
- [ ] Explained why each concept or property is essential for your program.

Program Development and Coding

- [ ] Finished coding the program.
- [ ] Debugged and tested the program.
- [ ] Checked if the program meets the design criteria.

Difficulties and Opportunities

- [ ] Described a difficulty faced or an opportunity realized in detail.
- [ ] Explained the steps taken to resolve the difficulty or seize the opportunity.

Finished Project Code Outline

- [ ] Listed only one instance of each important coding concept used.
- [ ] Provided the line numbers for each coding concept.
- [ ] Included a brief definition for each coding concept.
- [ ] Highlighted the aspect you are most proud of.

Future Goals

- [ ] Discussed what specific changes you'd make if given more time.
- [ ] Reflected on your approach and planning for the project.
- [ ] Considered resource utilization and time management.

Final Check

- [ ] Revisited the rubric to make sure you meet all the criteria.
- [ ] Made sure your document is well-organized and free of grammatical errors.
- [ ] Ready to submit the project and this document.

By ticking off each item on this checklist, you ensure that you have met all the requirements and can confidently submit your work for evaluation.